Digitising Fels Cave, Lelepa Island, Vanuatu

Nurmikko-Fuller, Terhi
terhi.nurmikko-fuller@anu.edu.au
Australian National University

Nelson, Kit
kit.nelson@anu.edu.au
Australian National University

Ballard, Chris
chris.ballard@anu.edu.au
Australian National University

Wilson, Meredith
memlwilson@gmail.com
Stepwise Heritage and Tourism Pty. Ltd

Lore Matanik Farenearu, Richard
rmatanik@gmail.com
Lelema World Heritage Committee

Willie, Edson
ewillie@vanuatu.gov.vu
Vanuatu Cultural Centre

Introduction

Fels or Feles Cave on Lelepa Island forms part of the UNESCO World Heritage cultural landscape of Chief Roi Mata's Domain in northwest Efate, Republic of Vanuatu (Wilson, Ballard and Kalotiti 2011). The Indigenous land-owning community has led the process of World Heritage inscription and site management from the start, including decisions around the management of Fels Cave. This large chamber cave, approximately 50m wide and 35m high (Fig. 1), is of considerable cultural and spiritual significance both locally and nationally through its association with sixteenth-century chief, Roi Mata, who is the focus of a mass burial on the nearshore island of Artok or Retoka, visible from the mouth of Fels Cave. The scale and shape of the cave and its fairly remote location, combined with an impressive range of painted and engraved rock art (Fig. 2) covering most of the 3000-year period of human settlement of the region, presents an interesting challenge for digital documentation. In this paper we focus on the photogrammetry workflow developed to capture the cave and its rock art as a 3D rendering.

Figure 1 Fels Cave (Chris Ballard, 2001)

Figure 2 Rock art of Fels Cave (Chris Ballard, 2005)

Methodology

Given the limited accessibility to the site, its isolation and limited transport infrastructure, SfM photogrammetry offered an ideal balance of durability of equipment, its compact size and weight, precision of data gathered and cost. Photogrammetry has already demonstrated its value in documenting cultural sites (Kowlessar et al. 2022; Brandolini et al. 2020; Cassidy et al. 2019;) The research team including members of the Lelema Community walked from Natapao village to the cave each day Data capture commenced with a Community member addressing the Cave and its spiritual occupants.

The data, consisting of a total of over 11,000 digital images taken over four days, was captured by the research team and members of the Lelema World Heritage Committee and the local Lelepa community, as well as staff from the Vanuatu Cultural Centre, using four different devices and two file types (Fig. 3). Settings on the cameras were adjusted to reduce sensor movement and to ensure low ISO. Where possible, a ring flash was used for illumination, except when using the Canon 700D, which had a built-in flash. Cameras were stabilised with a monopod for wide shots and handheld for closer details. The images were later merged using Agisoft's Metashape software on a custom-built processing PC with an AMD Ryzen 9 3900 12-Core 3.79GHz Processor, 128GB DDR4 RAM, and Nvidia GeForce GTX1660 6GB Graphics Card.

Camera	Lens	File format
iPhone 11 Pro	In-built camera and lens	HEIC
Canon 6D Mk1	EF50mm f/1.8 STM and EF100mm f/2.8 Macro USM	RAW format (CR2)
Nikon Z6	NIKKOR Z 24-70mm f/4.5	RAW format (NEF)
Canon 700D	EF-S18-55mm f/3.5-5.6 IS STM	RAW format (CR2)

Figure 3: Image capture equipment

Following the methodology first described by Mudge et al. (2010), each camera was used to capture parts of the cave in detail in image arrays. The image arrays all share the same properties: minimal focal adjustment, aperture size (f/), shutter speed, distance to subject. These settings changed depending on the lighting conditions and availability of accessible and stable terrain. Shots captured had 60% overlap within image arrays. Calibrated 12-bit target-marked scale bars were used to determine scale. Although the approach outlined

by Mudge et al. (2010) suggests a much lower tolerance, the range of devices used and the high variability of focal length, adjusted image brightness and ISO, combined to lift the tolerance to generate the minimum required number of points for a dense point-cloud and model.

The images were converted to lossless .TIF format prior to loading into Agisoft Metashape. The arrays were brought together and aligned with a larger general array of 1,164 images (captured using the Canon 6D Mk1) that were used to create a preliminary rendering of the bottom of the Cave. In total 7,733 digital images, translating to 7,382,566 tie points, were used to produce the initial sparse cloud (Fig. 4). A total processing time of approximately six days resulted in the production of the first draft of a polygon 3D model, consisting of 404,801,270 faces and being 38.1GB in size (Fig. 5).

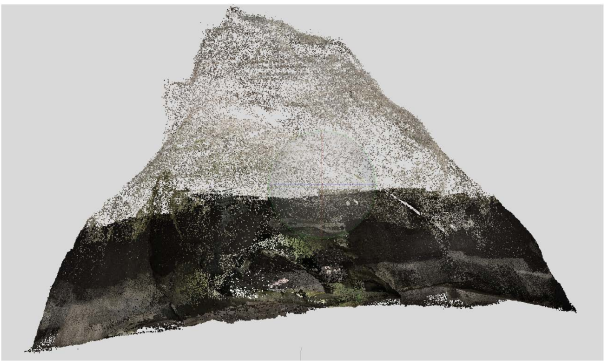

Figure 4 Initial sparse cloud generation demonstrating high number of tie points after optimisation

Figure 5 First draft of the polygon 3D model with backface culling (cave entrance at left)

Results and Discussion

The 3D virtual capture of Fels Cave serves a number of purposes: i) capturing and recording the cave's rock art, which spans approximately 3,000 years, from prehistoric engravings and paintings through to graffiti added between the 1870s and the present day; ii) creating a record of the current state of the cave structure, as a baseline against which changes such as additional graffiti or cave collapse can be monitored; iii) making the cave more accessible for those who may be unable to visit in person or in the event of closure of the cave due to instability or earthquake activity; and iv) as a way of disseminating information and increasing public awareness to new audiences, including the economically significant tourism industry. All participants were trained in basic photogrammetry as part of the project. This skills transfer, and a deliberate decision to minimise reliance on specialist equipment and to assess the suitability of (relatively) readily available hardware and software, illustrates the potential for community members to reproduce the workflow at a later date, here or at other locations of significance. One of the biggest challenges is the computational infrastructure needed for the processing of the images to create the 3D model - it is here that inter-institutional collaborations can support the project workflow. This is also true for the long-term storage of the 11,000 digital photographs and the hosting of the resulting 3D model. The copyright and intellectual property rights over the raw data and the model are held by the Lelema (Lelepa and Mangaliliu villages) community.

We acknowledge that, although an effort was made to capture the cave to scale and true colour, wear and changes over the four days of capture, our presence within the cave, choice of devices, and our abilities in capturing the rockwall make this 3D model a product of interpretation. As the cave was captured over four days, the model does not represent a single point in time. As practitioners we negotiate this temporal composite as a single authentic static moment in time constructed from our own experiences of being at Fels Cave. On 17 December 2024, just months after the field trip to record the cave, a major earthquake caused the cliff-face around the cave entrance to collapse; the cave is currently inaccessible, and the digital documentation has thus assumed a new significance as the primary record of the cave and its art, and a baseline for any future assessment of impact.

Bibliography

Enter your references here: Brandolini, Filippo / Cremaschi, Mauro / Mariani, Guido S. / Zerboni, Andrea / Lischi, Silvia / Degli Esposti, Michele (2020): "SfM-Photogrammetry for Fast Recording of Archaeological Features in Remote Areas", in: Archeologia e Calcolatori <https://hal.science/hal-03929093> [28.04.2025].

Cassidy, Brendan / Sim, Gavin / Robinson, David Wayne / Gandy, Devlin (2019): "A Virtual Reality Platform for Analyzing Remote Archaeological Sites", in Interacting with Computers 31: 167–176.

Kowlessar, Jarrad / Moffat, Ian / Wesley, Daryl / Jones, Tristen / Aubert, Maxime / Willis, Mark / Nayinggul, Alfred / the Njanjma Aboriginal Corporation (2022): "Applications of 3D Modelling of Rock Art Sites Using Ground-Based Photogrammetry: A Case Study from the Greater Red Lily Lagoon Area, Western Arnhem Land, Northern Austra-

lia”, in Visual Heritage: Digital Approaches in Heritage Science. Cham: Springer International Publishing: 93-114.

Mudge et al. (2010). Principles and practices of robust photography-based digital imaging techniques for museums. In VAST 2010: The 11th International Symposium on Virtual Reality, Archaeology and Cultural Heritage.

Mudge, Mark / Schroer, Carla / Earl, Graeme / Martinez, Kirk / Pagi, Hembo / Toler-Franklin, Corey / Rusinkiewicz, Szymon / Palma, Gianpaolo / Wachowiak, Melvin J. / Ashley, Michael / Matthews, Neffra / Noble, Tommy / Dellepiane, Matteo (2010): "Principles and Practices of Robust Photography-Based Digital Imaging Techniques for Museums“, in: VAST 2010 (ed.): The 11th International Symposium on Virtual Reality, Archaeology and Cultural Heritage.

Paris, France, September 2010: 1–27. <https://repository.si.edu/handle/10088/67710> [28.04.2025].

Wilson, Meredith / Chris Ballard / Douglas Kalotiti (2011): “Chief Roi Mata’s Domain: Challenges for a World Heritage Property in Vanuatu”, in Historic Environment 23, 2: 5–11